# Modelling

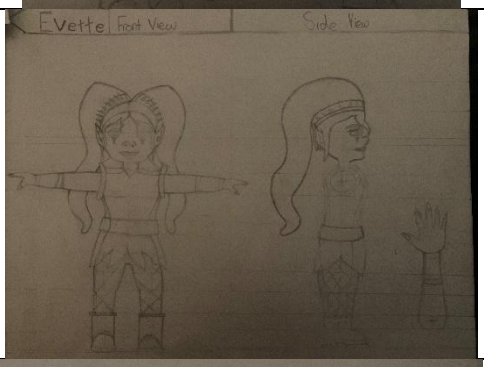
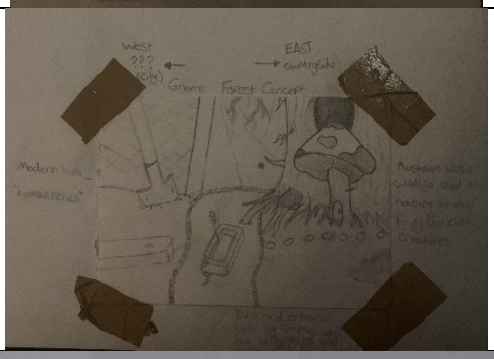
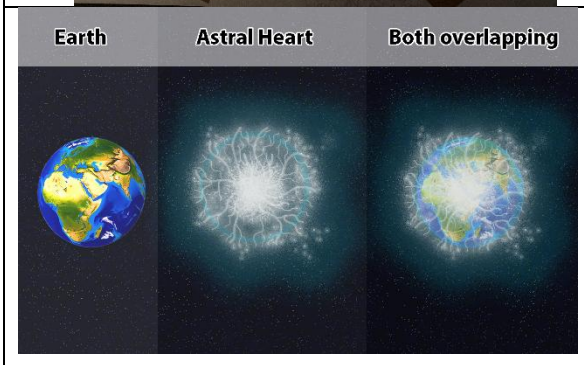
KIERAN SCHOFIELD

Character Design: 3
Character concept art 3
Character Mood board 4
Prop Design: 5
Prop concept art 5
Prop Mood board 6
Character model sheet 7
Technical character design notes 7
Prop model sheet 8
Technical prop design notes 8
Story 9
The World 9
The Gnome Village 9
Main Character 10
Other key characters in the story 10
Main Story 11
Storyboard 13
Technical modelling 14
Polycount, T-pose render & close-up. 14
UV maps 15
Reflection 16
What I enjoyed 16
Where I struggled 16
Useful tools 16
What I will do differently 16

Kieran Schofield

# Character Design:

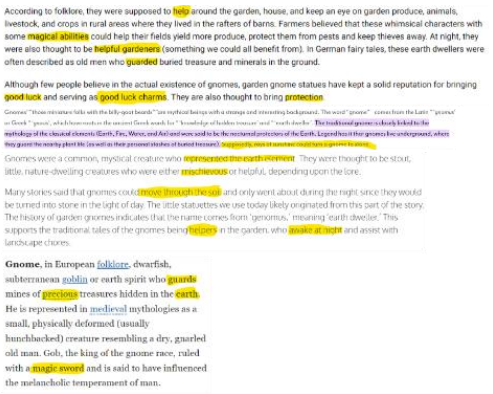
## Character concept art

| Art | Brief description |
|---|---|
|  | Original design concept – originally going for more of a 'gnome warrior' this was used as a basis to develop a character style. |
|  | Re-design of Evette concept – This used the theme and clothing aspects from the original but added a jester-like mixed with punk aesthetic to better reflect her personality visually. |
|  | World - Gnome Village concept art |
|  | World – Astral heart/world overlapping<br>(explained in lore section – created as a visualizer for drawings in the curator's room) |

Kieran Schofield

## Character Mood board

- *Top left is some folklore of gnomes from various websites.*
- *Bottom left are images of garden gnomes and ideas of their environment.*
- *Top right shows AI (Artificial Intelligence) generated concept images using the main theme words for my character 'gnome, jester, harlequin & fantasy' in assorted styles.*
- *Bottom right is harlequin & fantasy gnomes/art.*


According to folklore, they were supposed to help around the garden, house, and keep an eye on garden produce, animals, livestock, and crops in rural areas where they lived in the rafters of barns. Farmers believed that these whimsical characters with some magical abilities could help their fields yield more produce, protect them from pests and keep thieves away. At night, they were also thought to be helpful gardeners (something we could all benefit from). In German fairy tales, these earth dwellers were often described as old men who guarded buried treasure and minerals in the ground.

Although few people believe in the actual existence of gnomes, garden gnome statues have kept a solid reputation for bringing good luck and serving as good luck charms. They are also thought to bring protection.

Gnomes were a common, mystical creature who represented the earth element. They were thought to be stout, little, nature-dwelling creatures who were either mischievous or helpful, depending upon the lore.

Many stories said that gnomes could move through the soil and only went about during the night since they would be turned into stone in the light of day. The little statuettes we use today likely originated from this part of the story. The history of garden gnomes indicates that the name comes from 'genomus,' meaning 'earth dweller.' This supports the traditional tales of the gnomes being helpers in the garden, who awake at night and assist with landscape chores.

**Gnome**, in European folklore, dwarfish, subterranean goblin or earth spirit who guards mines of precious treasures hidden in the earth. He is represented in medieval mythologies as a small, physically deformed (usually hunchbacked) creature resembling a dry, gnarled old man. Gob, the king of the gnome race, ruled with a magic sword and is said to have influenced the melancholic temperament of man.











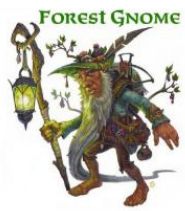








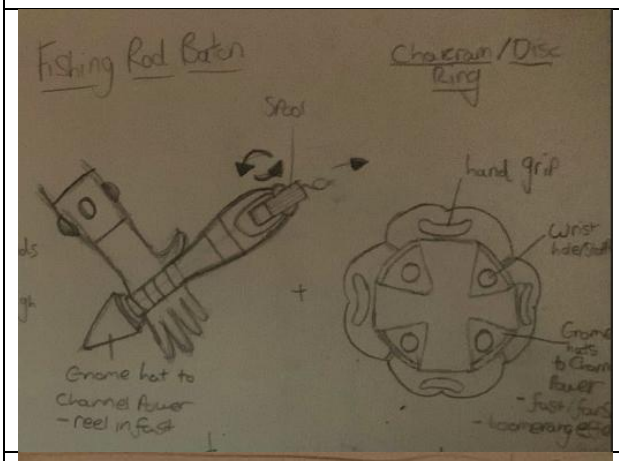
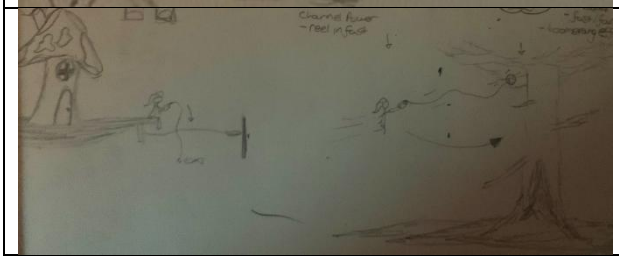







# Prop Design:

## Prop concept art

| Art | Brief description |
|---|---|
|  | Concept art for both the 'fishing baton' and chakram disc for Evette |
|  | Concept art of where the 'fishing baton' and chakram disc can be carried by Evette |
|  | Movement concept utilizing 'fishing baton' for fishing on left and being used with the disc to create allow for a swinging motion |

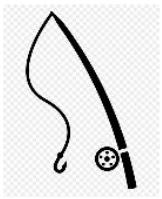
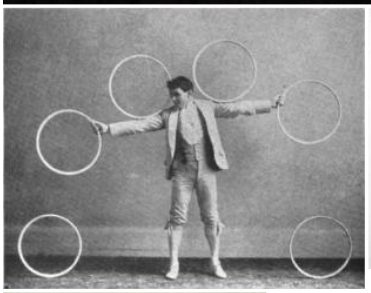
Kieran Schofield

## Prop Mood board

- *Top left shows batons, rings, and a pair of tonfa*
- *Top right shows various fishing rods from fantasy to real.*
- *Bottom shows characters using fishing rods for an idea of motion & scale – it also shows a character using 'ODM' gear from Attack on Titan for inspiration in how the prop can be used for movement.*







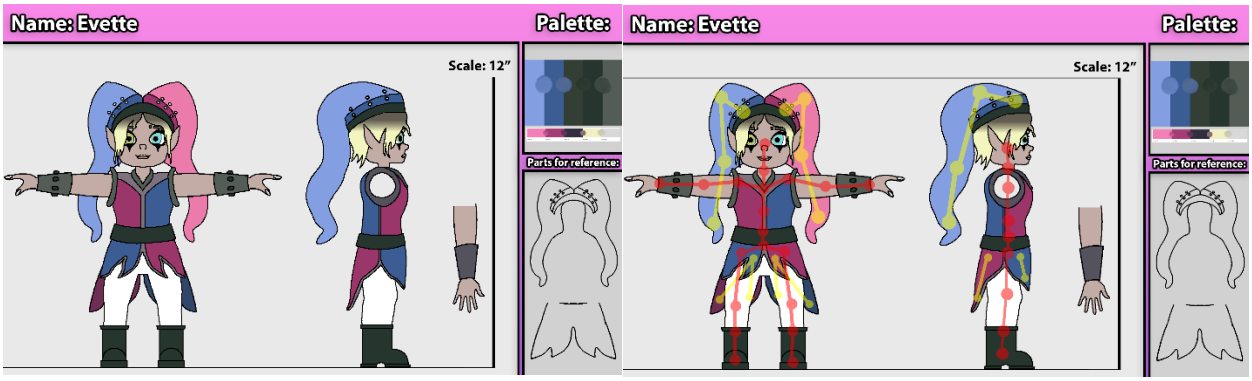




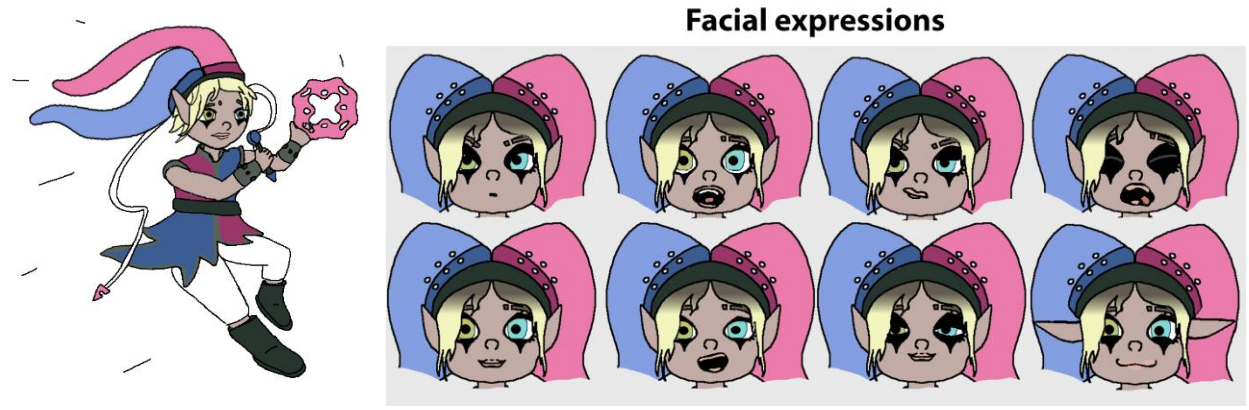


# Character model sheet

**Evette – Reference sheets + with ideal skeleton (Red for character movement & yellow for main cloth motion)**




**Evette – Action pose sketch and facial expressions**



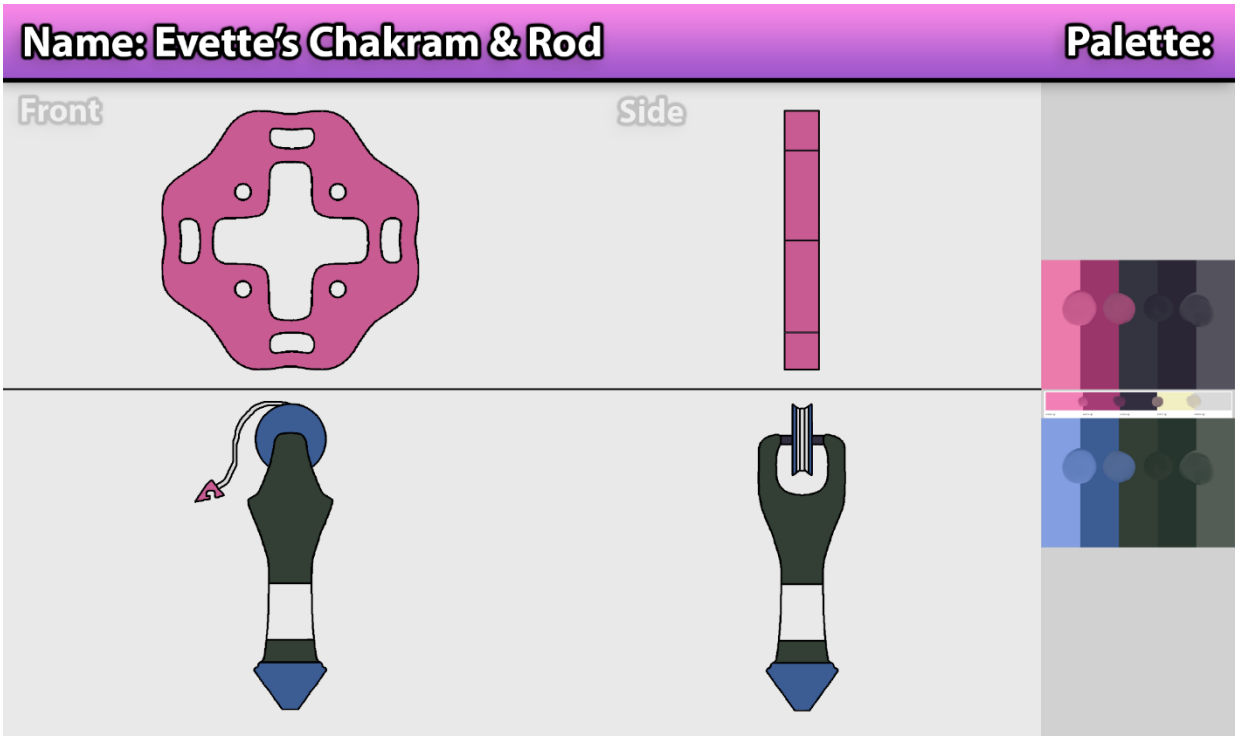

## Technical character design notes

**Animation style:** Stylized with realistic proportions and exaggerated features (big hat), fantasy elements in a Victorian era – like Dragon Prince & How to train your dragon. Similar in tone to Over the garden wall with friendly visuals that also work for a darker undertone.

**Potential issues:** The issue for animating and rigging this character from design is that there is potential for lots of clipping. Primarily between the hat and ears and between the waistline of the clothes and legs.

# Prop model sheet

**Evette's Chakram & Rod – Reference sheet**




## Technical prop design notes

**Animation:** Stylized with realistic applications, able to be utilized together and should be scaled so that the hook of the rod can fit into the holes of the chakram.

**Potential issues:** The main issue I think will be with the wire of Evette's rod being difficult to animate. It is very thin and will require a long ring or potentially blended shapes to unravel in a believable way. It will also be important to scale the hook correctly so that it can fit within the chakram's holes to be used as intended. It will also be important to use a simple texture for the wire as it will be susceptible to lots of deformation with the way it will bend and curve around.

# Story

## The World

The world the story takes place in is like our own in all aspects apart from the Astral heart. This resides inside the planet and is unseen to the naked human eye. It functions like a human heart and distributes astral energy from all areas of the world to pump the energy into 'empty' areas known as hotspots.

Hotspots are areas where life used to be and no longer is, anything left in the area with enough depth for a history/story/purpose will have astral energy sent to it to help continue the cycle of life. (Varying from things as small as children's toys and dolls to larger mythological creatures from statues/fallen kingdoms.

History etc. also dictates how much astral energy is sent to the creatures) Astral energy is what gave the initial spark of life to this version of Earth and is how humans came to live and adapt over the years. The heart's main function is to create life and it does it unbiasedly and will not harm or hinder any life it has created.

When a creature passes away it is returned to the astral heart through its vein-like network, that is spread through the planet and out into the atmosphere, to be sent elsewhere on the network to create new life.

Humans make up most of known life on the planet. Due to the heart distributing the energy to only 'empty' areas the newer creatures and life forms the heart creates are never seen by humans and often dismissed as fairy tales/conspiracies.

## The Gnome Village

The Gnome Village was the result of a prior hotspot. An old, abandoned garden that was filled with astral energy leading to the inanimate gnomes left there to spring to life. The culture and typical characteristics of the new life created are initially based upon the folktales or history known in the world about the creature, these can change over time but initially all creatures follow what is already known.

In the case of the gnome village, they were given life and became a tribe of intelligent guardians and gardeners that help local/countryside dwelling humans. The culture being intelligent allowed them to develop over time and adapt to new surroundings the same as humans have and live peacefully hidden from the human population.

The culture taking years to adapt is also why characters like Evette can be so different from the initial rigidness of their initial culture.

Gnome hats are what allow them to tunnel underground - allowing them to create tunnels or swim through the earth by spinning themselves at high speeds.

## Main Character

**Name:** Evette - Name meaning, evokes both the Old World and the New. A variant of the French Yvette, meaning “yew,” Evette is also an unusual Americanization of the name Eve. Since in the Abrahamic religions Eve was the first woman and the mother of all people, Evette means “living one.”

**Character traits:** Mischievous, curious, creative tinkerer, skilled, athletic, and untraditional to gnome values. Evette’s initial character starts with her being torn between her own curiosity and confidence and the values of her village and father being imposed on her.

She has a unique movement style; using her baton to swing herself around the forest instead of tunneling like the other gnomes.

Daughter of the 'head hat' - maker of the gnome hats, key role in their society who is close friends with the chief of the village.

**Appearance:** 'Uniform' (Folklore gnome attire) is customized in a harlequin style two-tone fashion; her hat originally is a very standard pointy gnome hat her father made for her. Her outfit is split into pink and blue to highlight her louder confident nature and show how she does not fit in visually at first when compared to the other gnomes or her own hat. This changes as the story progresses with Evette receiving a hat with matching colors that ‘fixes’ her visuals and shows visually how her character has developed.

### Other key characters in the story

**Name:** The Curator

The main antagonist. A leading conspiracy theorist, collecting various creatures and mapping out the hotspots. Introduction scene with him displays his maps/private collections/etc. in a well-kept but scrambled room.

**Appearance:** The man is tall and slender, wearing a deep red suit with every edge coming to a point - menacing silhouette. He also wears a large red crooked 'top hat' - looking more like a witch's hat with all the pointed edges. His face is also pointy, a curved pointed nose, sharp jaw and hunter eyes which are obscured by his small 'Pince-nez' style tinted glasses. His private team also all wear various red outfits fitting for their professions within the museum staffing. The main theming of the design is loud and overly pointed/designed to contrast the simplicity of Alexander in both character and visuals.

**Name:** Alexander (Alex) Name meaning, Greek — Defender of men and In Biblical naming the meaning is: One who assists men.

A young human tailor assistant who assists Evette when her hat is split in two. Alexander is young, polite, unjudging, calm, and kind. is meant to reflect the unbiased view, imagination, and sense of wonder we all have as children before we begin to entirely understand the way the world works.

**Appearance:** Overall simple and peaceful, beige Victorian era clothing with a matching oversized bakers/news style cap. All well-kept and clean. His physical features are also very plain and earthy, brown hair, brown eyes, and a short stature due to being young. His clothing is well fitted due to his profession but has entirely soft edges.

## Main Story

1. The story begins with Evette, alone making a mess as she zooms through the gnome village swinging with her baton & disc until she arrives at her class.

*(The class is small; only four with Evette included. Only one of the three acknowledges her while the others shun her.*

2. The leader of the small group explains that they will be going on an excursion to the west rather than their usual route east.

*(The usual route east leads to rural/countryside towns that are much tamer and accustomed to the folklore of the gnomes being helpers, whereas the west leads to a 'new' city - unbeknownst to the gnome village.)*

3. The leader explains that not all the members will be going on venture as it is unknown to them, the only person excluded from going is Evette due to her reckless nature.
4. Evette sneaks out - following the group all the way to the city.

*(The gnomes are naturally surprised at the strange environment and are overwhelmed and separated due to the high traffic along the roads.)*

5. Evette follows the group she is spotted and in attempt to flee gets her hat caught in a wire fence.
6. Calling out to the group for help, her friend from the class rushes back, yanking Evette free but splitting her hat down the middle in the process.
7. The pair attempt to escape together but they are once again separated by a building crowd of people with big clunky Victorian flash cameras.

*(Her friends hat falling in the process leaving her unable to flee.)*

8. Evette manages to evade the crowd in a similar fashion to the way she zoomed around the village and makes her way to the group hiding in a nearby park.

*(The group is sad but the leader acts coldly, noticing her damaged hat he knew Evette would not be able to tunnel to hide or quickly maneuver the new terrain.)*

9. The leader demanded Evette return to the village while the rest of the group looked for the missing gnome.
10. Grabbing her split hat, Evette runs away from the group.

*(Distraught but determined to help her friend.)*

11. Evette ran her way through the quieter streets of the city and bumped into Alexander's leg who was walking out from a nearby fabric shop.
12. Alexander apologizes and helps Evette up with just one finger before offering to repair her hat.

*(Alexander was on his way home - left by his mother to lock up the shop for the night but noticing Evette's split hat Alexander offers to repair it. The is not surprised by Evette while Evette is the opposite, surprised, her overconfident nature slowly subsiding as she reluctantly grabs his finger for help.)*

13. Alexander repairs Evette's hat, sewing fabric sides onto the sturdy base, creating two points where one used to be jester-like in appearance.

14. Evette uses her new hat to glide around the room but bangs into something - Alexander catching her.

*(The pair converse as he repairs her hat and the moments building to the new hat show a building of trust in others and between their species human/gnome and a development to Evette's character in being able to accept help from others. It also helps develop a friendly trust between the two as they explain their own interpretation of their different 'worlds.')*

15. The pair prepare to find Evette's friend and begin to discuss plans, mapping out potential areas for her friend in crude crayon drawings.
16. Meanwhile the curator has already found and captured her and is attempting to get information from the friend.
17. The man makes many vague threats, using clever wordplay to eventually get the friend to reveal there are more gnomes in the city.

*(Evette and Alexander are following their crudely drawn map when they hear talk of a 'grand reveal' at the museum tonight. The curator is planning to display the gnome to the citizens to provide proof to his ongoing theories.)*

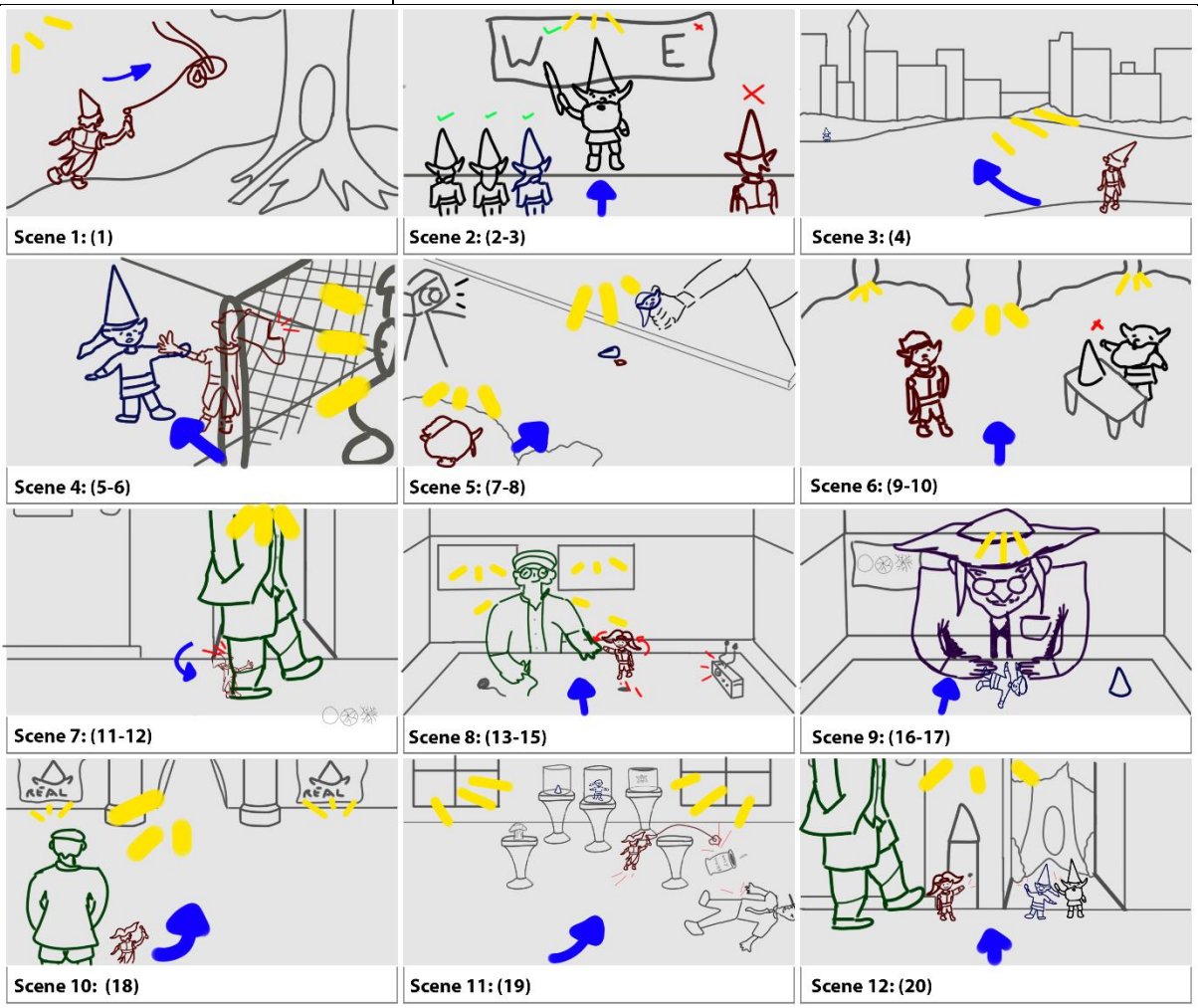
18. Evette and Alexander make their way to the museum to save her friend while curator's team of red suits clumsily work together to catch them.

*(The red team are entirely disorganized and uncaring, some leaving instead of following the curators' orders which leads to the team slowly falling apart through miscommunication. This helps to show how important teamwork is and when compared to Evette and Alexander helps display that people can work together even if they are different.)*

19. The pair triumphant in finding the lost friend just before the curator can show the world her locked up and the curator is then seen as a fraud for having an empty cage and the two make their escape.
20. Meeting back up with the other gnomes, Evette decides to stay with Alexander to learn the curator's reasoning for kidnapping her friend and to act as a scout for the 'new world.'

## Storyboard

| Key | |
|---|---|
| **Lighting** | Yellow |
| **Camera/Focus motion** | Blue arrows |
| **Interaction/Indicator** | Red/Green lines |
| **Scene (number)** | Correlates to the number in the main story section |



# Technical modelling

## Polycount, T-pose render & close-up.

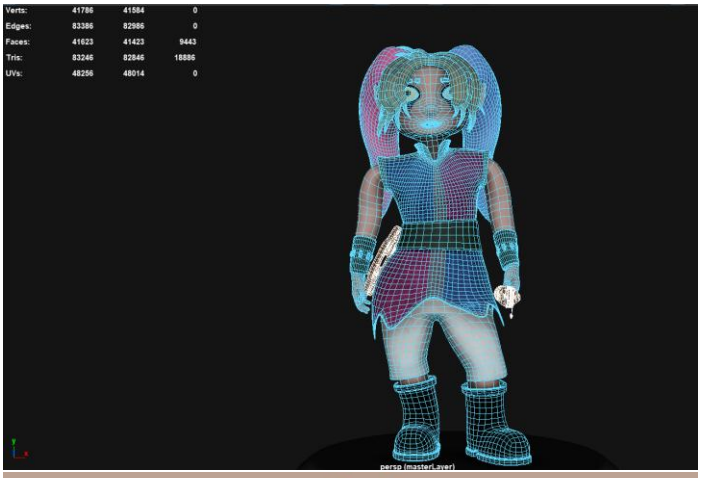
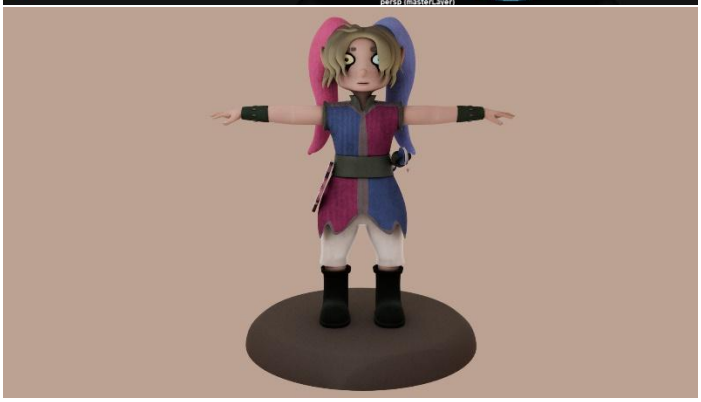



Image displaying the polycount of the entire model.



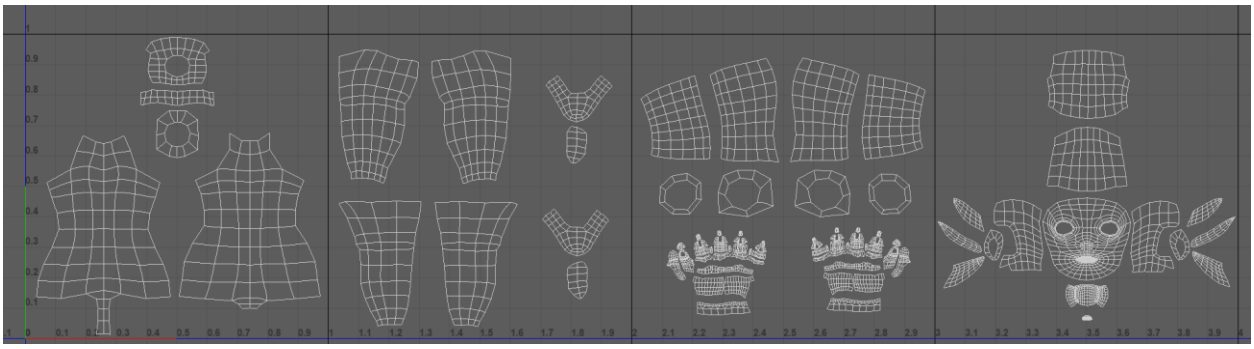
Rendered image showing the working textures in the T pose. (Posed version removes height map.)



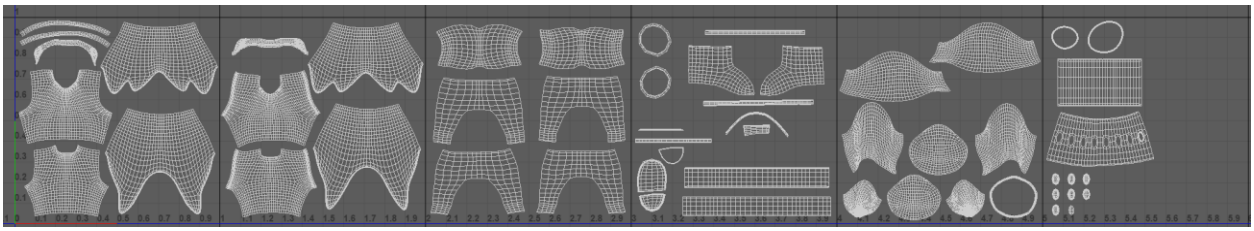
Close up on Evette, highlights other areas for potential deformation around the armpit edges of the clothing.

## UV maps

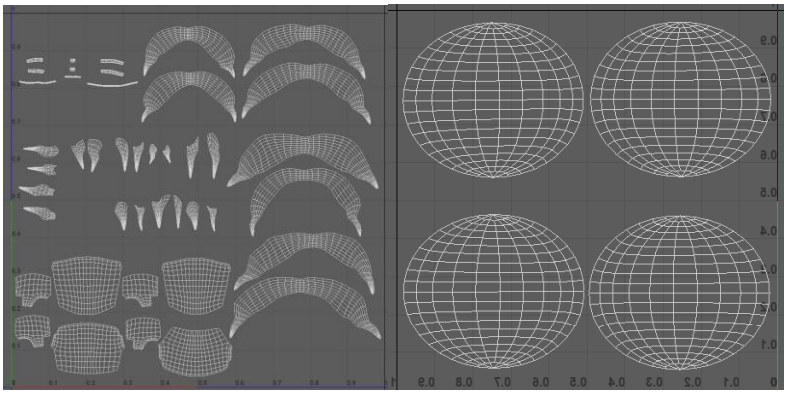
**UVS for the body (Body, legs/feet, arms/hands and head)**



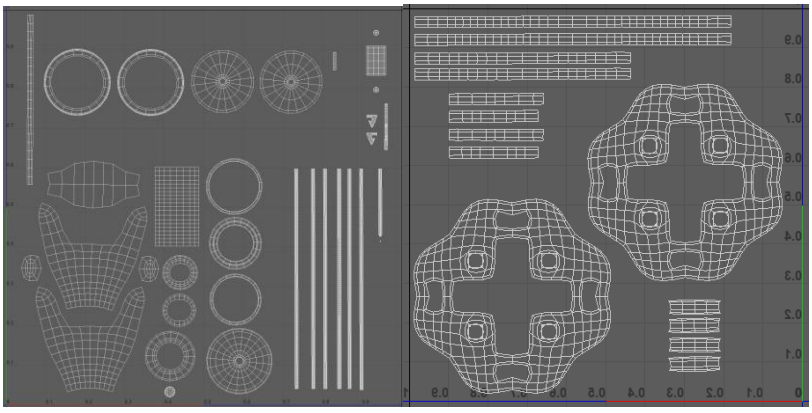
**UVS for the clothing (Outer shirt, inner shirt, bottom wear, shoes, hat and wristbands)**



**Hair & Eyes**



**Fishing Rod & Chakram (Props)**



# Reflection

## What I enjoyed

I really enjoyed seeing my character come to life from drawing to 3d and seeing the modelling progress from a cube to something that was recognizable as a character. I spent lots of time researching and finding references to create a story that I felt could be both cartoonish but still believable for the setting and overall feel happy with the character that I have created conceptually.

## Where I struggled

The main area I struggled with was translating the 2d design to 3d. I struggled with this mostly due to my drawings not being the best and originally did not matching up how I wanted them to. This made the progress quite slow, and I had to redraw parts like the hat, eyes, and face to better help myself find a balance. There are still aspects and proportions in the initial design like the arm size and head shape that I slightly changed the scale of to better suit 3d space.

I also had some difficulties with getting the textures to look correct in the rendering with the custom pose. I am unsure why the Mixamo rigged model will not recognize the UDIMS and tried different methods to fix it and re-assigning the materials but sadly did not work. (This is also why I have rendered still images of the t-pose version to highlight the textures.)

## Useful tools

The most useful tools I found for this in Maya were the soft select and mirror tool. This allowed me to make small adjustments to the topology to make things smoother without having to manually drag each vertex (most of the time). The mirror tool made creating the symmetry of the body and face much easier and if I made a mistake on one side, I would be able to delete and mirror it over.

Outside of Maya, the Moodle resources really helped, especially when it came to rendering the different passes. A YouTube channel and website named Academic Phoenix plus was also useful for the short guides and had interesting maya plugins that create an auto rig from Mixamo. Sadly, I did not get a chance to properly use this but will try it out in the future. (https://www.academicphoenixplus.com)

## What I will do differently

In the future I will make a few more concept sketches and have close ups with more detail for certain body parts and areas of the character. This is to make sure I do not have to spend as much time going back and forth and can get things right in the initial pre-production phase.

I will also try to manage my workflow better as I left the texturing till quite late into production and did not have enough time to really push the stylized look that I wanted for the character. I would also assign the materials different for a faster workflow from substance to Maya. While I am still happy with the results, I would like to attempt to create a more stylized painted texture and learn how to use the AI toon shader in Arnold to get a nice 2d/3d look. I would also like to spend time finishing the alternate clothing I had created which shows Evette in her outfit before the main story.

I would have also liked to create a more dynamic pose however the Mixamo rigs tended to stretch either the hat or the clothing and clipped quite a lot with itself – once I have learnt rigging, I would like to repose the character to closer match the drawing I made in the concept phase.